

Putting Patterns, Data and Points Together

Answer Key

Grades 4–5 Mathematics

Quick Answers

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|---|--|
| 1. 31 | 8. 6 |
| 2. 43 | 9. 2 |
| 3. 2 | 10. (a) y-coordinate for term 4 = 15, — |
| 4. 16 | 11. (a) median number of books = 3.5, (b) mean number of books = 3.5 |
| 5. 4 | 12. (a) books in week 6 = 23, (b) books over the first six weeks = 78, — |
| 6. x-coordinate: pattern S term 5 = 19, y-coordinate: pattern T term 5 = 25 | |
| 7. 42 | |
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What is verified

10 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 10, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grades 4–5 Mathematics

5.G.A.1–A.2 – *problems 6, 10*

5.MD – *problems 5, 7, 8, 9, 11, 12*

5.OA.B.3 – *problems 1, 2, 3, 4, 12*

Difficulty 1–5 of 5 across 17 tagged checks.

- 1.** Write the 8th term of pattern S (start at 3, add 4).

Solution: The 8th term is 7 steps past the start: $3 + 4 \times 7 = 3 + 28$. Extending the table confirms it: 3, 7, 11, 15, 19, 23, 27, 31. 31

- 2.** Write the 8th term of pattern T (start at 1, add 6).

Solution: Seven steps again, larger step: $1 + 6 \times 7 = 1 + 42$. Extending: 1, 7, 13, 19, 25, 31, 37, 43. 43

- 3.** At which term number do patterns S and T give the same value?

Solution: Read straight down the two rows of the table and look for a column where the entries match. Term 1 gives 3 and 1; term 2 gives 7 and 7.

Algebraically, $3 + 4(n - 1) = 1 + 6(n - 1)$ gives $2 = 2(n - 1)$, so $n - 1 = 1$.

T starts 2 behind and gains 2 each step, so it draws level after exactly one step.

2

4. How much greater is pattern T's 10th term than pattern S's 10th term?

Solution: Nine steps past each start. T: $1 + 6 \times 9 = 55$. S: $3 + 4 \times 9 = 39$. Subtract: $55 - 39$.

16

5. How many students read exactly 3 books?

Solution: Find the tick labelled 3 on the Book Plot and count the crosses stacked above it.

4

6. Write the ordered pair (pattern S's 5th term, pattern T's 5th term).

Solution: Both 5th terms are 4 steps past their start. S: $3 + 4 \times 4 = 19$, which the question puts first, so it is the across number. T: $1 + 6 \times 4 = 25$, the up number.

(19, 25)

7. How many books did the 12 students read altogether?

Solution: Multiply each book count by the number of students at that tick, then add:

$$2 \times 2 + 4 \times 3 + 4 \times 4 + 2 \times 5 = 4 + 12 + 16 + 10$$

42

The count of crosses (12) and the total of books (42) are different questions about the same plot — reading the wrong one is the usual slip here.

8. How many students read more than 3 books?

Solution: “More than 3” leaves out the stack sitting exactly at 3. Count the stacks to its right: 4 students at 4 books and 2 at 5 books, so $4 + 2$.

6

9. How many more students read 3 books than read 5 books?

Solution: Compare stack heights, not book counts: the stack at 3 is 4 crosses tall and the stack at 5 is 2 crosses tall, so $4 - 2$.

2

10. (a) Write the y -coordinate of the ordered pair for term 4 of pattern S. (b) Plot the pairs for terms 1 to 4.

Solution (a): Term 4 is 3 steps past the start: $3 + 4 \times 3$.

15

(b) — instructor-judged (a drawing). The four pairs are (1, 3), (2, 7), (3, 11) and (4, 15): term number across, pattern value up.

Check that each mark sits one square right and four squares up from the one before, and that all four fall on a straight line. A constant-step pattern always plots straight, so a bent line means a term was generated wrongly.

11. (a) Median number of books read. (b) Mean number of books read.

Solution (a): In order, the 12 values are 2, 2, 3, 3, 3, 3, 4, 4, 4, 4, 5, 5. With an even count the median sits between the 6th and the 7th: those are 3 and 4, so the median is halfway between them. 3.5

(b) Total books divided by number of students: $42 \div 12$. 3.5

The two agree because this plot is symmetric — two crosses at each end, four at each middle value.

12. A club's weekly totals follow pattern S. (a) Books in week 6. (b) Books over the first 6 weeks. (c) How could the mean check part (b)?

Solution (a): Five steps past the start: $3 + 4 \times 5$. 23

(b) Add the six weekly totals: $3 + 7 + 11 + 15 + 19 + 23$. Pairing from the ends helps — $3 + 23 = 26$, $7 + 19 = 26$, $11 + 15 = 26$, so three 26s. 78

(c) — instructor-judged. Look for the mean-times-count idea. The mean of the six totals is 13, and a mean multiplied by how many values it came from must give back the total: $13 \times 6 = 78$. That is a genuine check because it reaches the same number by a different route. A response that just adds the six numbers again has repeated the work rather than checked it.